

Afro-centred Collaborative care: Technology support for Type 2 Diabetes Management in Port Harcourt, Nigeria

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ABSTRACT

Type 2 Diabetes (T2D) poses a significant health challenge in Nigeria, requiring long-term community care involving diverse stakeholders. The condition leads to severe health complications exacerbated by suboptimal lifestyle conditions. However, sparse knowledge exists regarding the role of technology in facilitating collaborative T2D care in Nigeria. Additionally, contextual factors, including socio-cultural and demographic influences on the use and adoption of existing T2D technologies, remain poorly understood. Diverse collaborative care approaches involving community pharmacists and caregivers, and context-specific design implications for effective T2D care are also insufficiently documented. Addressing these knowledge gaps, my research project adopts a decolonial Afro-centred perspective to probe the landscape of T2D care involving multiple stakeholders in Port Harcourt, Nigeria. Also, I aim to improve research methodologies and design practices for sustained community care, with a focus on co-designing collaborative T2D care interventions.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Collaborative care; Community care; Digital Health; Decolonization; Type 2 Diabetes

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1 CONTEXT AND MOTIVATION

Nigeria confronts a rising prevalence of Type 2 Diabetes (T2D) and other non-communicable diseases (NCD) [19]. The International Diabetes Federation (IDF) projects a significant rise in the estimated T2D cases in Nigeria, forecasting an increase from 3.6 million in 2021 to nearly 5 million by 2045 [8]. It is imperative to acknowledge, however, the inherent inconsistency in data pertaining to T2D

prevalence in Nigeria. Various research papers present disparate figures, ranging from 5.7% [11] to 4.3% [38], and even exhibiting significant variability, extending up to 11% [36]. The significant absence of reliable data is equally evident in Port Harcourt - the research context. Recognized as a prominent developing Nigerian city, Port Harcourt bears a substantial burden of T2D in Nigeria [37].

Within the Nigerian healthcare landscape, effectively managing NCDs such as Type 2 Diabetes requires a long-term collaborative synergy among healthcare professionals, complementary care providers, and caregivers [33–35]. However, we have observed a notable gap in understanding the active involvement of stakeholders beyond physicians, such as community pharmacists, in T2D care within this context. Despite the existence of numerous digital health interventions for T2D care [10, 21, 26, 30], a dearth of research exists concerning the influence of socio-demographic factors on the collaborative use of technology for T2D care and context-specific digital requirements for fostering supportive and collaborative T2D care. Additionally, a substantial portion of existing digital health interventions in this research context is not tailored to meet the specific needs of its target population [12, 31, 32]. This has given rise to a prevalent “one-size-fits-all” approach characterized by the top-down transfer of interventions from global northern regions to their southern counterparts, exemplified here by Nigeria [17, 25, 29]. Recognizing these inherent limitations, my research aims to fill this gap through the design and implementation of a mixed-methods study using a decolonial Afro-centred perspective as supported by HCI researchers in this field [2, 16].

2 RESEARCH QUESTIONS AND OBJECTIVES

My research focuses on investigating the collaborative care roles of people with Type 2 Diabetes (PWT2D), community pharmacists, and caregivers. It specifically examines their use and appropriation of existing technologies and explores their perspectives on emerging technologies. This will provide a broad understanding of the demographic and socio-cultural context of pharmaceutical and technology enabled collaborative care, thereby informing future designs.

Three overarching questions reflect the intention of this study, including:

- (1) **Question 1:** How is technology currently being used in supportive and collaborative T2D care in Port Harcourt, and what are the socio-cultural and demographic factors influencing them?
- (2) **Question 2:** What are the design implications of this understanding for emerging technologies suitable for this context?

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- (3) **Question 3:** Using this project as a case study, how can we conduct research and co-design workshops remotely in global southern community contexts?

To answer the stated research questions, my thesis adopts a hybrid approach, involving digital and traditional quantitative and qualitative methods of data collection. I also intend to explore remote co-design practices using speculative approaches facilitated by WhatsApp, design probes and other context-specific toolkits [13, 20, 40]. Preliminary research suggested a lack of sufficient peer-reviewed literature on remote co-design activities before the COVID-19 pandemic [1, 22, 27]. However, since the onset of the pandemic, numerous studies leveraging communication platforms like Zoom or Microsoft Teams have emerged in both the Human-Computer Interaction (HCI) and Computer-Supported Cooperative Work (CSCW) fields [24, 28, 40]. Notably, co-design activities involving WhatsApp use remain significantly underexplored.

The ensuing theoretical framework of my study, Afro-postmodernism, is decolonial in nature and a modification of the prevailing Afrocentric paradigm used in Human-Computer Interaction (HCI) for development research [39]. This approach and perspective signifies a decolonized paradigm characterized by the unequivocal rejection of racism and exclusionary elements. It encourages the pragmatic acceptance of certain non-African concepts, which, despite their colonial origins, are assessed primarily on their practical efficacy rather than the historical motivations underpinning their existence.

Considering the research motivation and questions, my study aims to achieve the following objectives:

- (1) **Objective 1:** To generate an understanding of the intricate nature of informal and community-based self and collaborative T2D care
- (2) **Objective 2:** To generate scalable and extrapolatable design implications for context-specific collaborative T2D care.
- (3) **Objective 3:** To explore the feasibility of Afro-postmodernism as a decolonial praxis for digital health research in low resourced regions.
- (4) **Objective 4:** To explore the use of both digital (i.e., WhatsApp) and traditional methods of data collection and to make suitable recommendations for exploring research contexts using accessible communication technologies like WhatsApp.
- (5) **Objective 5:** To develop and investigate new and understudied approaches for conducting co-design activities on virtual communication spaces, like WhatsApp.

3 POSITIONALITY STATEMENT

As an immigrant residing in the United Kingdom, people from other nationalities have asked me to describe unique aspects of my culture and how health practices are defined within it. It breaks my heart to say that while I am an advocate of culture, I do not know a tenth about my ethnicity from lived experience. However, I have had the opportunity to witness, observe, and learn about multiple cultures, their health-seeking behaviors, and general attitudes towards health through my involvement in public engagement activities and my work as a clinical and, thereafter, community pharmacist. These events have significantly influenced my positionality.

I grew up, lived, and worked in a city that although was less than 197 kilometers from my village, was cosmopolitan by Nigerian standards. As a post-colonial young millennial, my attitudes towards technology, health and design practices differ significantly from my Generation X parents. My genealogy has been hugely impacted by colonial education that any rationality of thought towards Afrocentricity or other decolonial concepts could be perceived as some sort of rebellion by my progenitors. For these reasons, I am motivated to explore the nuances of context-inspired community care. This exploration considers core aspects that promote my inclusion and representation as a researcher while challenging existing paradigms for research in this context. Simultaneously, I aim to investigate opportunities for technology use and design, ensuring relevance to a diverse audience, including people like myself. Through this process, I will attempt to make the concept of decolonization tangible by proposing frameworks that may facilitate research and design practices for managing long-term health conditions in both in-person and virtual spaces.

4 RESEARCH METHODS

My study is divided into three phases, including the information gathering and contextualization phase, co-design phase, and in-the-wild evaluation phase.

- (1) **Phase 1:** This phase has been recently concluded. It was a mixed methods study that involved qualitative and quantitative data collection using both digital and traditional methods of data collection. After reviewing secondary literature, holding recurring meetings with supervisors, and obtaining ethics approval from the University of Port Harcourt in Nigeria, the data collection process commenced. It involved in-person and synchronous WhatsApp interviews, automated WhatsApp surveys, and concluded with the distribution of questionnaires in Port Harcourt. Synchronous WhatsApp and in-person interviews actively engaged members PWT2D, community pharmacists, and caregivers) of The Diabetes Care Network (TDCN)- a Port Harcourt-based organization providing diabetes-related health education and digital healthcare assistance to PWT2D and their caregivers. Several other participants were contacted through non-governmental organizations and religious institutions using snowballing techniques. On the other hand, to obtain quantitative data, WhatsApp was automated using a cloud communications platform called Twilio. This was inspired by Fei et al.'s work [23] on automated WhatsApp questionnaires in panel surveys. Automated surveys on WhatsApp relied significantly on PWT2D who had expressed interest following a targeted social media campaign, while physical questionnaires were distributed to PWT2D using guerrilla techniques, a traditional method of data collection in this research context.
- (2) **Phase 2:** Building upon the design opportunities identified in Phase 1, such as the notable absence of context-inspired digital interventions for collaborative T2D care, this second phase is dedicated to using a decolonial Afro-centred perspective for co-designing an intervention that supports and facilitates collaborative T2D care. I intend to conduct virtual

co-design workshops, guided by principles and approaches that worked in the first stage of the study, such as using instant messengers e.g., WhatsApp for co-design workshops and following the Afro-postmodernist framework for virtual co-design. I also intend to use Afro-centred community inspired design approaches [9, 14, 15] in a virtual co-design space. Excluding the mobile version of WhatsApp, tentative platforms to be used include the mobile version of Miro (an online whiteboard platform) and Post-it (mobile sticky notes). Participants will have to choose and/or suggest what platform works best for them. If none of the tentative platforms proves feasible, I intend, through a collaborative approach, to explore ways of using the WhatsApp virtual space to facilitate co-design activities.

- (3) **Phase 3:** This phase is significantly dependent on the outcomes of phase 2. I intend to conduct iterative tests in situ and explore the potential for the intervention to be applied in similar global southern contexts, not only for T2D management but also for the collaborative care of other long-term health conditions.

5 CURRENT RESULTS AND EXPECTED CONTRIBUTIONS

The first phase of the study generated unique insights that have fed into academic contributions and workshops, some of which have been peer-reviewed and others which are still in-the-works.

5.1 Insights from Phase 1

- (1) The qualitative study shed light on the contextual role of caregivers and community pharmacists in collaborative T2D care, as well as people with T2D in self-care. PWT2D experienced varying levels of informal, community, and pharmaceutical care; nevertheless, their willingness to engage in collaboration with their children and pharmacists within a digital space was somewhat constrained by a myriad of socio-demographic and contextual factors. This phase of the study also unveiled the significant role of religion and complementary care providers, such as Traditional Medicine Practitioners (TMPs), in the provision of T2D care.
- (2) Results obtained from the quantitative study offer statistical insights into the epidemiology of the disease, socio-demographic factors influencing T2D management, and additional design opportunities for using technology in T2D care.
- (3) Opportunities for design include involving community pharmacists in shaping interventions like mobile apps, promoting patient communication and pharmaceutical care. This should be done with due consideration for the socio-financial landscape of Nigeria, where many interventions clash with global payment and usage standards. Furthermore, there is potential in developing and implementing digital health policies and frameworks, aiming not only to reconcile the recognised tension between pharmacists and physicians but also to dispel misconceptions about pharmacists and highlight their role in community care. Another avenue involves fostering the ethical use of OpenAI tools [18] by pertinent

stakeholders to gather context-specific health information. Additionally, designers can explore adding features that facilitate the monitored use of herbal products for chronic disease care.

5.2 Expected contributions

- (1) Phase 1 of the study yielded three empirical contributions: a paper exploring the complexities of Type 2 Diabetes community care among caregivers, pharmacists, and people with T2D; a statistical contribution on the socio-demographic influences of T2D; and a reflective contribution that exhaustively describes the methods of data collection using WhatsApp and how the obtained data was analyzed. These unpublished contributions all have design implications that are relevant to Human-Computer Interaction (HCI), Computer-Supported Cooperative Work (CSCW) and Digital Health fields.
- (2) An opinion piece crafted as a scoping review of HCI academic contributions on decolonial and Afrocentric approaches for design and research [5]. This paper questions the positionality of researchers in the Human-Computer Interaction for Development (HCI4D) fields, challenges some underpinning notions of the leading paradigm for Afrocentred research by Asante [7] -Afrocentricity- and proposes a decolonial praxis for Afro-centred research.

5.3 Dissertation status and next steps

The Centre for Doctoral Training (CDT) in Digital Health and Care at the University of Bristol, is a full-time four-year program that provides a unique foundation in the dynamic world of digital health. I am currently in my third year, which included an intensive taught year and one year dedicated to the first phase of my research. The second phase of my study is projected to conclude in my third year. The initial months of my fourth and final year will be dedicated to the evaluation phase, with the remaining months leading up to graduation focused on writing my dissertation chapters. As part of a "Digital Health Promotion for Fitness Enthusiasts in Africa" [6]. Since then, I have made several poster presentations, organized a workshop at CSCW's 2023 Companion conference [3], and co-authored a short paper on the use of social media and instant messengers for collaborative T2D care in Port Harcourt [4]. I participated in the African Co-Design Academy co-located with **AfriCHI** in South Africa and the remote doctoral consortium which was organised a few days before the conference. The remote doctoral consortium had about four students in attendance, and a panel of three HCI academics.

Anticipated challenges in my study, which may persist in future phases, include the scarcity of peer-reviewed literature on collaborative care using supportive technologies for T2D in Nigeria and the use of WhatsApp as a synchronous research and co-design tool, remote recruitment of diverse stakeholders with minimal attrition rates, and the absence of nuanced frameworks for Afro-centred decolonial Human-Computer Interaction (HCI) research for managing long-term health conditions in Western Africa. Additionally, my background in pharmaceutical sciences and management influences my perspective, and unveils new challenges, as I attempt to carve a

niche within the Human-Computer Interaction for Health space. Through the doctoral consortium, I intend to learn as much as possible about navigating these challenges as an early-career researcher, especially in the absence of sufficient literature for reference and mentors with a similar academic background.

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